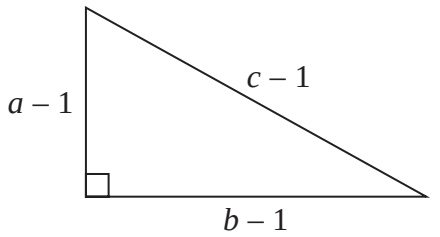
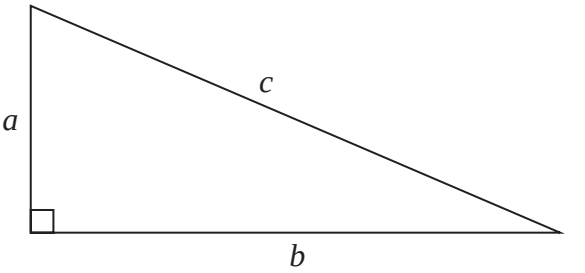
17 Here are two right-angled triangles.
All the measurements are given in centimetres.





Show that $2a + 2b - 2c = 1$

$a^2 + b^2 = c^2$

1 mark
Pythagoras on one triangle.

$(a - 1)^2 + (b - 1)^2 = (c - 1)^2$

$a^2 - 2a + 1 + b^2 - 2b + 1$

$= c^2 - 2c + 1$

1 mark
Use both triangles; expand every square.

Subtract $a^2 + b^2 = c^2$ from this:

$-2a - 2b + 2 = -2c + 1$

$2a + 2b - 2c = 1$, as required.

1 mark
Cancel equal squared terms, then rearrange.

(Total for Question 17 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA